WHAT MAKES SHORT FORM CONTENT GO VIRAL?



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AGENDA

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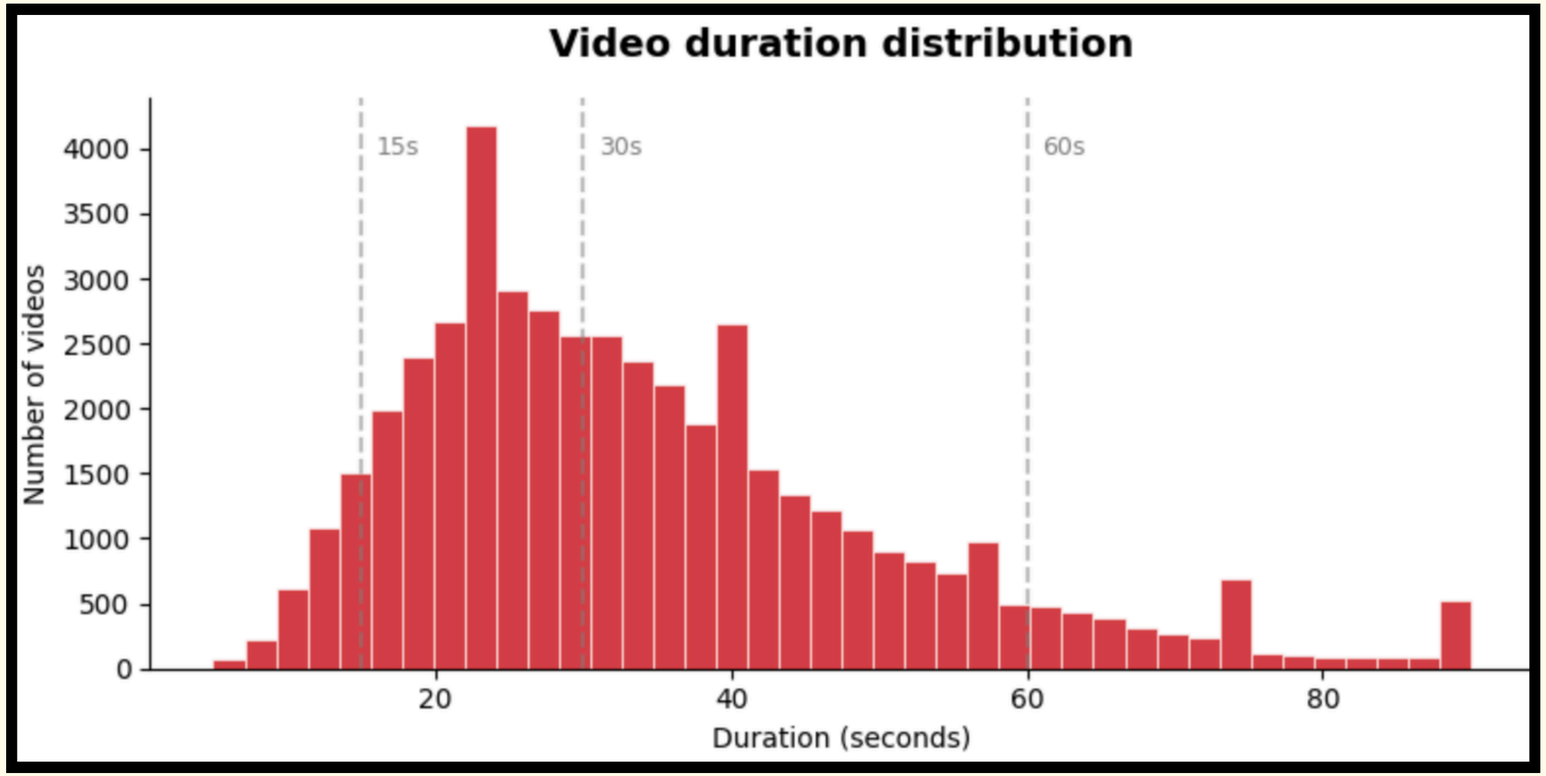
To help answer our main question, we've put together 7 different visualizations comparing different categories, like average views by upload hour, and compared them to see what actually makes short-form content go viral.

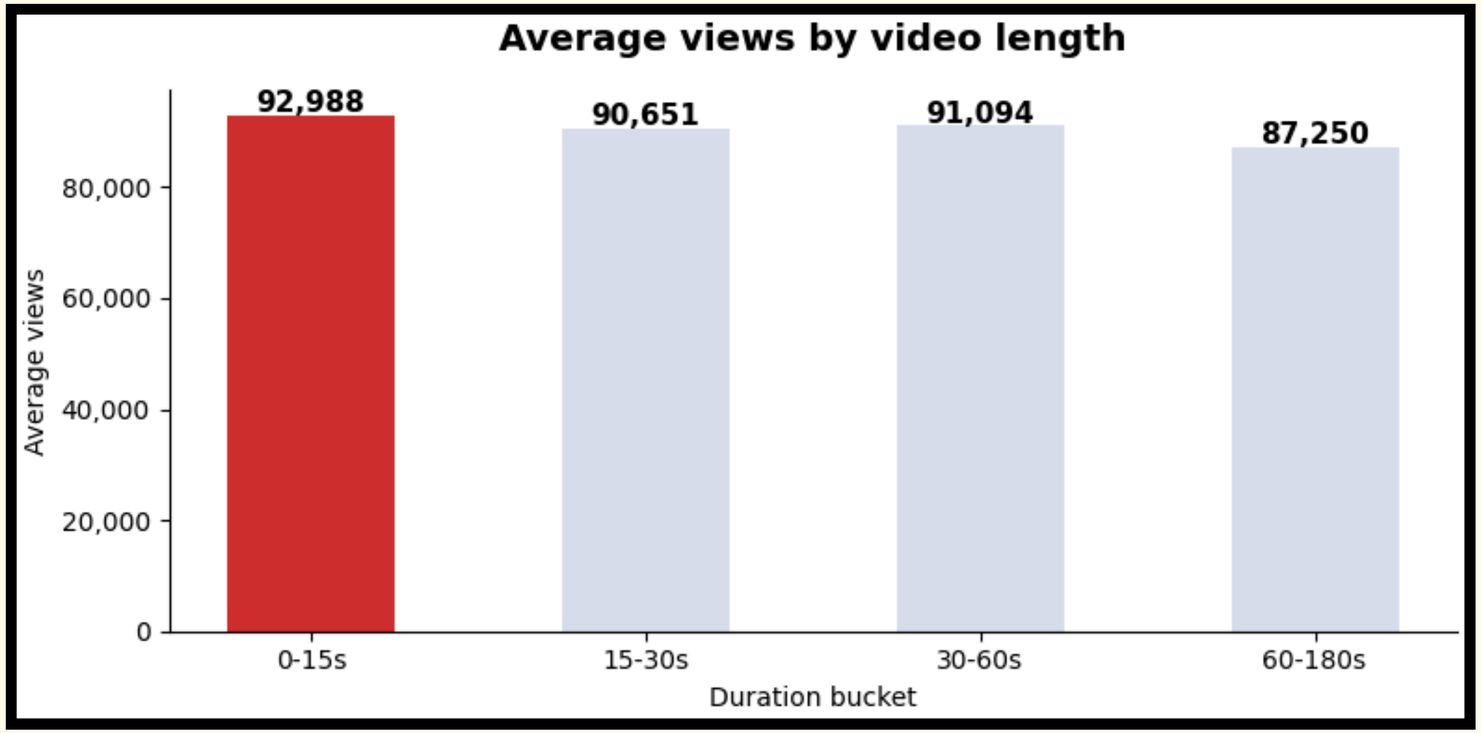
VISUALIZATIONS

DISTRIBUTION OF VIDEO DURATIONS

Top Findings:

- The distribution of the video durations is right-skewed.
- Sharp rise approaching 15 seconds duration, not as many videos beneath it.
- Sharp decline after the 60s, not many videos are greater than it.
- Finding: Most frequent video durations are within the 20-25s bucket. Small spikes at 40s, indicating that some longer style videos are also potentially popular.





AVERAGE VIEWS BY VIDEO LENGTH

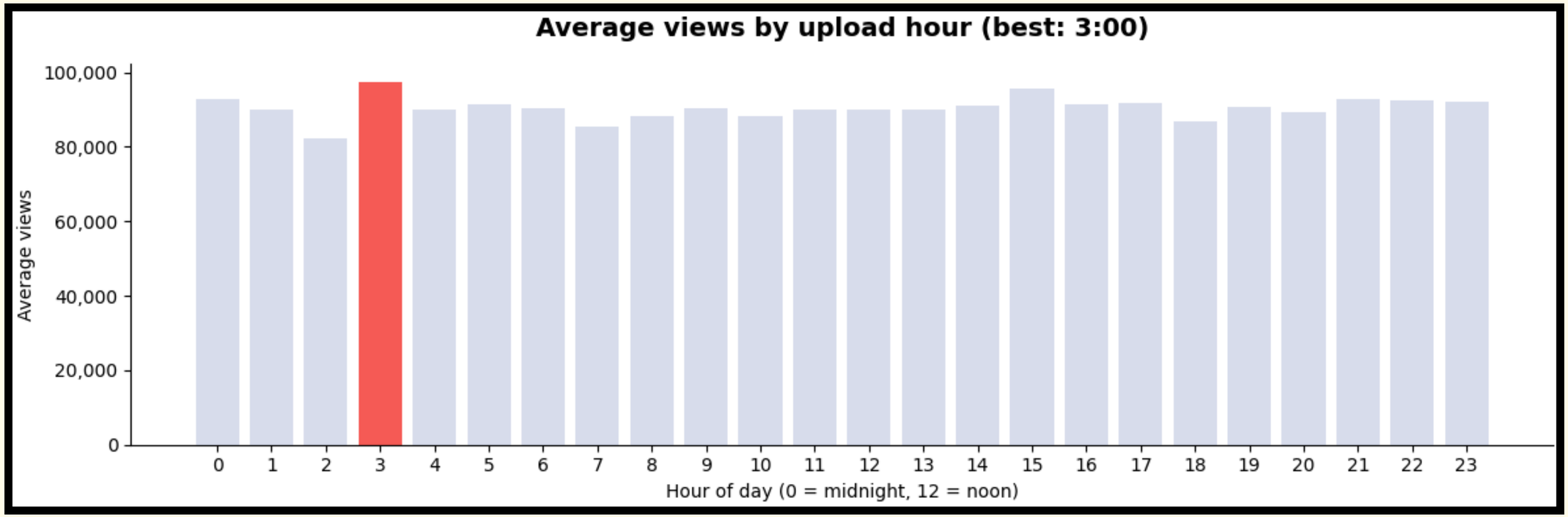
Top Findings

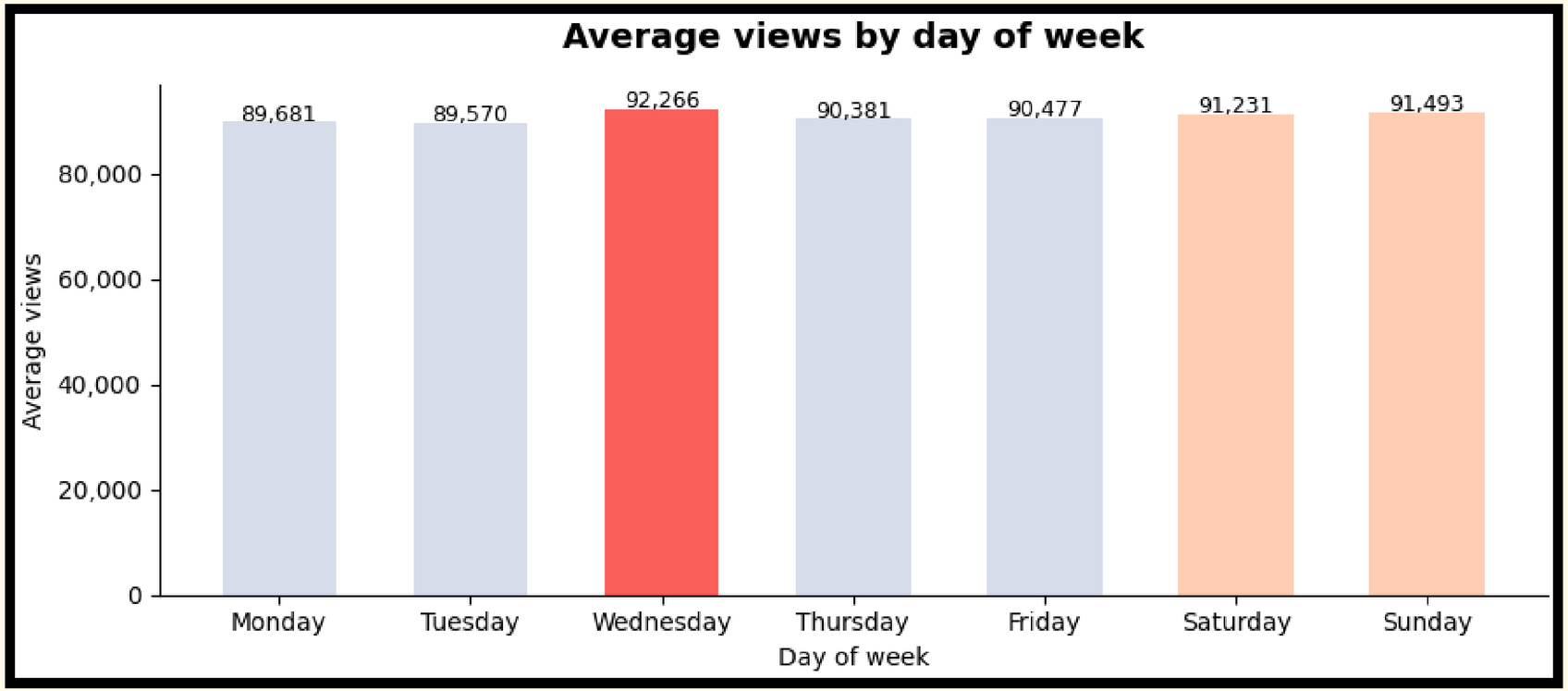
- Best video length: 0-15 seconds with avg 92,988 views
- Worst video length: 60-180 seconds with avg 87,250 views
- Difference between best and worst video length: 5,738 views
- Finding: The 0-15s bucket gets the highest average views, but the difference between buckets is small, shorter videos have a slight edge.

AVERAGE VIEWS BY UPLOAD HOUR

Top Findings

- Best hour: 3:00 with avg 97,509 views
- Worst hour: 2:00 with avg 82,414 views
- Difference between best and worst hour: 15,094 views
- Finding: Hour 3 has the highest average views, but the spread across hours is small, suggesting timing matters less than content quality.
- Why? The 3:00 AM peak likely captures a captive audience of insomniacs, night-shifters, or international viewers actively browsing.

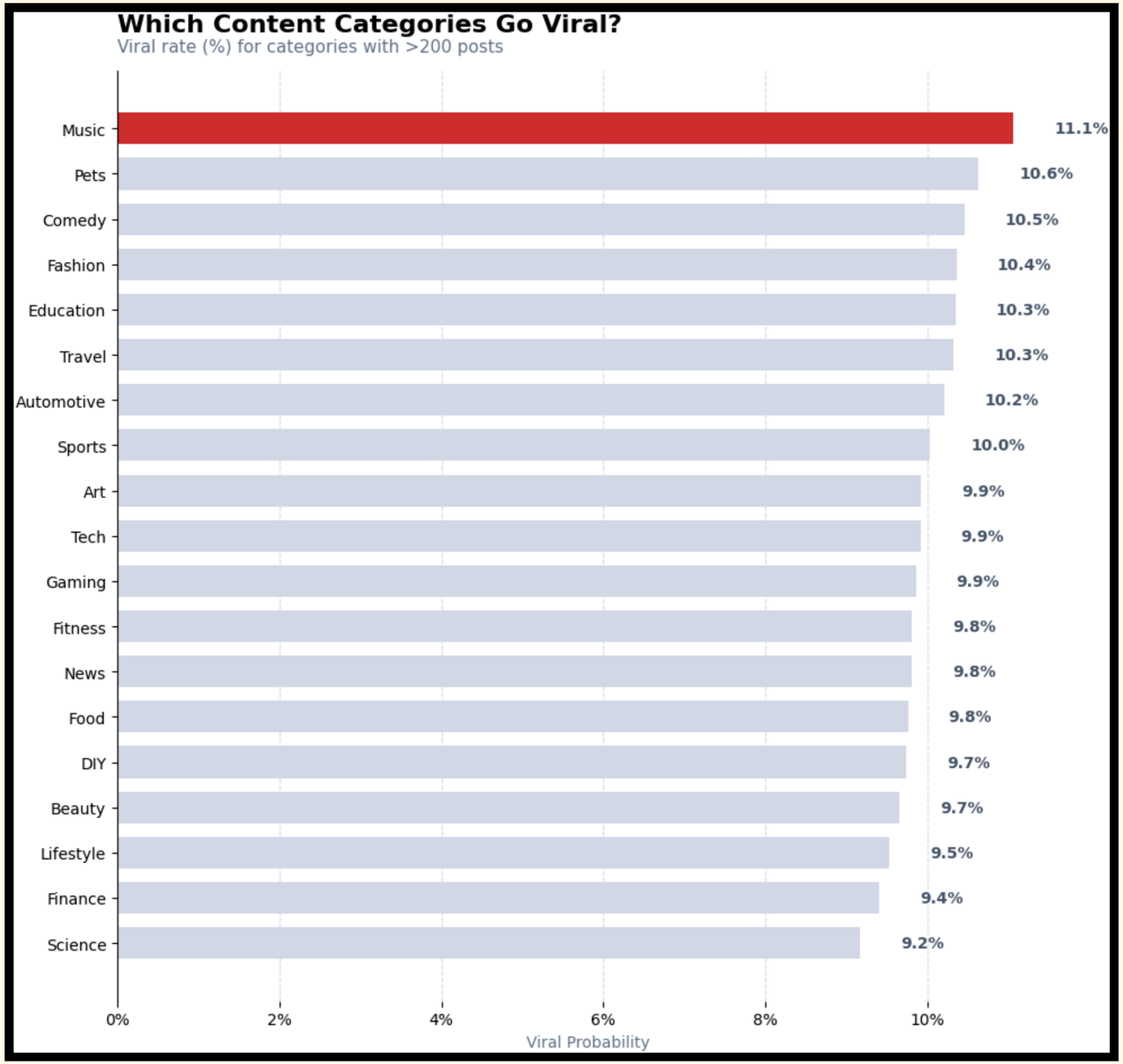




AVERAGE VIEWS BY DAY OF WEEK

Top Findings

- Weekend avg: 90,889 views
- Weekday avg: 90,486 views
- Finding: Wednesday has the highest average views, and weekends vs weekdays show a difference of 404 views on average.
- What does this mean? The marginal view difference between weekends and weekdays shows that modern users scroll consistently throughout the week, meaning "weekend free time" no longer creates a massive traffic spike over daily weekday habits.



VIRAL RATE BY CONTENT CATEGORY

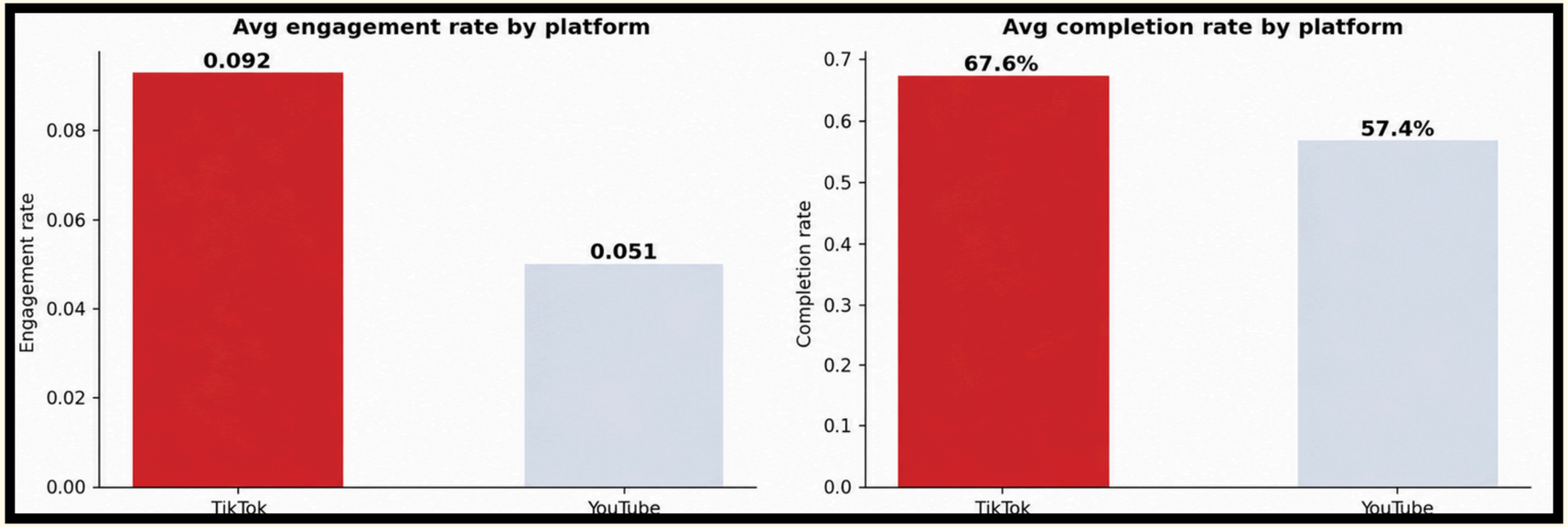
Top Findings

- Best Content Category: Music (11.1%)
- Content focused on immediate emotional or visual appeal like Pets (10.6%), Comedy (10.5%), and Fashion (10.4%) outperforms "serious" topics like Science (9.2%)
- Overall: very narrow margin between most categories, so category isn’t the only factor that drives success

ENGAGEMENT RATE BY PLATFORM

Top Findings

- TikTok has a significantly higher avg. engagement rate (0.092) compared to YouTube (0.051), showing stronger audience interaction.
- TikTok also leads in avg completion rate (67.6%) versus YouTube (57.4%), indicating viewers are more likely to watch content through to the end.
- TikTok outperforms YouTube across both engagement and completion metrics, suggesting it is the stronger platform for retaining viewer attention and driving interaction.



VIRALITY SCORE PREDICTOR

For our final project, we built a virality score predictor for YouTube videos. So, how does it work?

UI/UX:

UI (User interface) shapes the layout and interactive elements of the predictor using elements like input fields and buttons.
UX (user experience) helps to take in the user's URL input and display a result instantly.

- Used HTML to build the interface
  - — a container used to group other elements
  - <input> — used to get text from the user
  - <button> — a clickable button
  - style="..." — used to define CSS for an element
  - id="..." — gives a unique name to elements

- Used CSS to specify the style
  - background — background color
  - color — text color
  - padding — space inside an element
  - border-radius — rounded corners
  - font-size / font-weight — text size and boldness
  - display: flex — puts elements side by side
  - display: none — hides an element

```css
/* Button */
.analyze-btn {
  width: 100%;
  padding: 14px;
  background: var(--teal);
  color: #050709;
  border: none;
  border-radius: 12px;
  font-family: "Syne", sans-serif;
  font-size: 15px;
  font-weight: 700;
  cursor: pointer;
  letter-spacing: .04em;
  transition: background .15s, transform .1s;
}
.analyze-btn:hover  { background: #34d9c3; }
.analyze-btn:active { transform: scale(.985); }
```

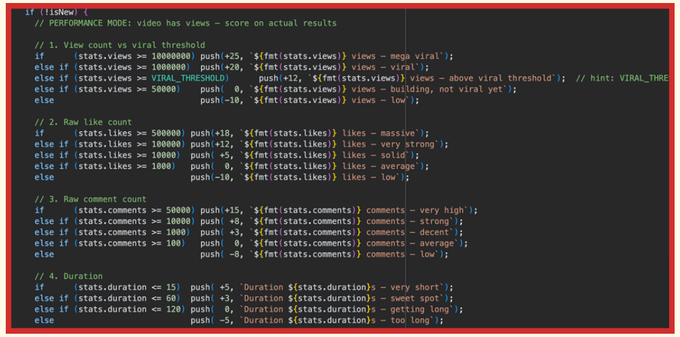
Setting the style

SCORING

How do we determine if a video will go viral?

Setting Benchmarks

```
// Benchmarks from our 47,598-video dataset
const VIRAL_THRESHOLD  = 20678;    // top 10% of views — hint: 206748
const NEW_VIDEO_CUTOFF = 50000;    // below this = treat as new — hint: 50000
const BEST_HOURS = new Set([3, 15, 5, 0, 22]);  // hint: 3, 15, 5, 0, 22
const BEST_DAYS  = new Set(["Wednesday", "Saturday", "Thursday"]);          // hint: "Wednesday", "Saturday", "Thursday"
```



```
if (!isNew) {
  // PERFORMANCE MODE: video has views — score on actual results

  // 1. View count vs viral threshold
  if      (stats.views >= 10000000) push(+25, `${fmt(stats.views)} views — mega viral`);
  else if (stats.views >= 1000000)  push(+20, `${fmt(stats.views)} views — viral`);
  else if (stats.views >= VIRAL_THRESHOLD)      push(+12, `${fmt(stats.views)} views — above viral threshold`);  // hint: VIRAL_THRE
  else if (stats.views >= 50000)    push(  0, `${fmt(stats.views)} views — building, not viral yet`);
  else                              push(-10, `${fmt(stats.views)} views — low`);

  // 2. Raw like count
  if      (stats.likes >= 500000) push(+18, `${fmt(stats.likes)} likes — massive`);
  else if (stats.likes >= 100000) push(+12, `${fmt(stats.likes)} likes — very strong`);
  else if (stats.likes >= 10000)  push( +5, `${fmt(stats.likes)} likes — solid`);
  else if (stats.likes >= 1000)   push(  0, `${fmt(stats.likes)} likes — average`);
  else                            push(-10, `${fmt(stats.likes)} likes — low`);

  // 3. Raw comment count
  if      (stats.comments >= 50000) push(+15, `${fmt(stats.comments)} comments — very high`);
  else if (stats.comments >= 10000) push( +8, `${fmt(stats.comments)} comments — strong`);
  else if (stats.comments >= 1000)  push( +3, `${fmt(stats.comments)} comments — decent`);
  else if (stats.comments >= 100)   push(  0, `${fmt(stats.comments)} comments — average`);
  else                              push( -8, `${fmt(stats.comments)} comments — low`);

  // 4. Duration
  if      (stats.duration <= 15)  push( +5, `Duration ${stats.duration}s — very short`);
  else if (stats.duration <= 60)  push( +3, `Duration ${stats.duration}s — sweet spot`);
  else if (stats.duration <= 120) push(  0, `Duration ${stats.duration}s — getting long`);
  else                            push( -5, `Duration ${stats.duration}s — too long`);
```

Determining Virality



```
} else {
  // PREDICTIVE MODE: new video — score on observable signals

  // 1. Duration
  if      (stats.duration <= 15) push(+15, `Duration ${stats.duration}s — very short, high completion likely`);  // hint: 15
  else if (stats.duration <= 60) push(+10, `Duration ${stats.duration}s — in the sweet spot`);                  // hint: 60
  else if (stats.duration <= 120) push(  0, `Duration ${stats.duration}s — getting long`);                      // hint: 120
  else                            push(-15, `Duration ${stats.duration}s — too long for a Short`);

  // 2. Upload hour — 3am, 3pm, 5am, midnight, 10pm perform best in our data
  if (BEST_HOURS.has(stats.uploadHour)) push(+12, `Posted at ${stats.uploadHour}:00 — one of the best hours`);
  else                                  push( -5, `Posted at ${stats.uploadHour}:00 — off-peak timing`);

  // 3. Day of week — Wednesday, Saturday, Thursday perform best
  if (BEST_DAYS.has(stats.uploadDay)) push(+8, `Posted on ${stats.uploadDay} — a strong day in our dataset`);  // hint: BEST_DAYS
  else                                push( 0, `Posted on ${stats.uploadDay} — average day`);

  // 4. Hashtag count — 5 to 15 is optimal
  if      (stats.hashtagCount >= 5 && stats.hashtagCount <= 15) push(+10, `${stats.hashtagCount} hashtags — well optimized`);
  else if (stats.hashtagCount >= 1 && stats.hashtagCount < 5)   push( +4, `${stats.hashtagCount} hashtag(s) — could use more`);
  else if (stats.hashtagCount > 15)                             push( -3, `${stats.hashtagCount} hashtags — possibly over-tagged`);
  else                                                          push(-10, `No hashtags — low discoverability`);

  // 5. Title length — 15 to 60 chars is optimal
  const tl = stats.titleLength;
  if      (tl >= 15 && tl <= 60) push( +8, `Title length ${tl} chars — optimal range`);
  else if (tl > 60)              push( -5, `Title length ${tl} chars — too long`);
  else                           push( -5, `Title length ${tl} chars — too short`);

  // 6. Category — top performing: Music(10), Pets(22), Fitness(17), Travel(19), DIY(26)
  const BEST_CATS = new Set([10, 22, 17, 19, 26, 1, 27]);
  if (BEST_CATS.has(stats.categoryId)) push(+7, `Category is top-performing in our dataset`);  // hint: BEST_CATS
  else                                 push( 0, `Category is average in our dataset`);
```

Predicting Virality

Determining Virality

- Determined benchmarks for videos to be considered “viral.”
- Based on the video’s views, likes, comments, and duration, we added to the video’s “score.”
- If the video’s score surpassed the “viral” threshold, then it would be considered viral.

Predicting Virality

- If a video is considered “new”, with under 50,000 views, we can predict whether it will go viral.
- Based on the video's duration, upload hour, hashtag count, upload day, title length, and category, we assign points once again to its score.
- From our exploratory data analysis, we found the best options for all these variables.
- If the score is high enough, the video will most likely go viral!

API

A score is given to ANY Youtube link pasted.

How???

What We Built:

- extractVideoId( ) – parses YouTube URLs in 3 formats (youtu.be, /watch, /shorts) using regex patterns.
- fetchStats( ) – async call to Youtube Data API v3 that returns 9 fields per video: title, views, likes, comments, duration, upload hour, upload day, title length, hashtag count and category ID
- analyze( ) – the orchestrator bound to the Analyze button: URL → video ID → API call → Group B's scoring function → result display.
- Built-in error handling for invalid URLs, missing videos, and failed API requests; supports keyboard Enter key for submission.

NO API = NO DATA = NOTHING TO SCORE

Using an API key...

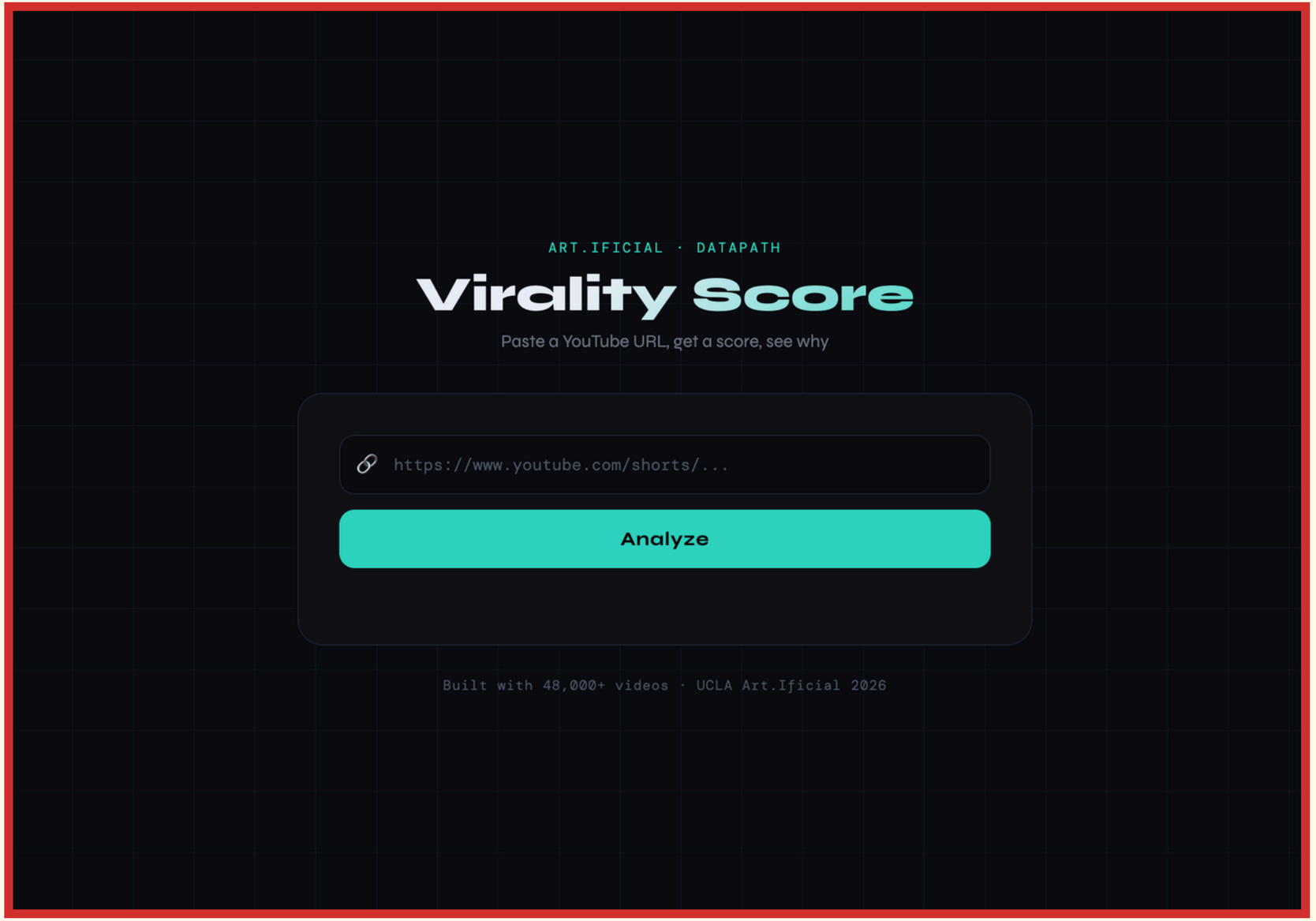
```
// ── API FETCH ──────────────────────────────────────────────────────
async function fetchStats(videoId) {
  const url  = `https://www.googleapis.com/youtube/v3/videos?part=statistics,contentDetail
  const res  = await fetch(url);
  const data = await res.json();

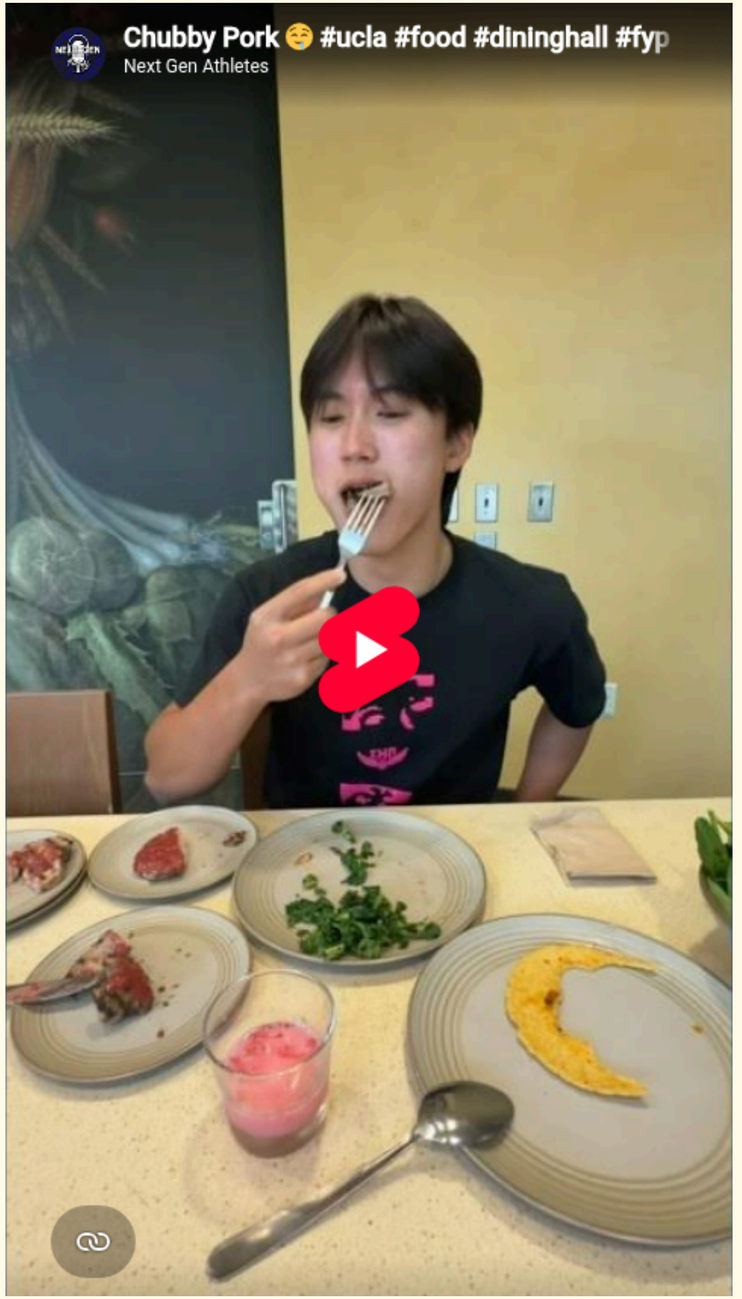
  if (!data.items || data.items.length === 0) throw new Error("Video not found");

  const item        = data.items[0];
  const snippet     = item.snippet;
  const stats       = item.statistics;
  const content     = item.contentDetails;
  const publishedAt = snippet.publishedAt;

  return {
    title:        snippet.title,
    views:        parseInt(stats.viewCount    || 0),
    likes:        parseInt(stats.likeCount    || 0),
    comments:     parseInt(stats.commentCount || 0),
    duration:     parseDuration(content.duration),
    uploadHour:   new Date(publishedAt).getUTCHours(),
    uploadDay:    getDayName(publishedAt),
    titleLength:  snippet.title.length,
    hashtagCount: countHashtags(snippet.description) + countHashtags(snippet.title),
    categoryId:   parseInt(snippet.categoryId || 0),
  };
}
```

Video





https://notnavv.github.io/Projects-Spring-2026/

THANK YOU!